

VENKATA KRISHNA SAADHVIK MUDDANA

513-349-6965 | muddanvk@mail.uc.edu | [LinkedIn](#) | [Github](#) | Open to Relocation

SUMMARY

Data Scientist and MS Computer Science candidate who builds predictive models, production ML pipelines, and stakeholder-ready analytics from large, multi-source datasets. Strong in Python, SQL, R, scikit-learn, and PyTorch, with hands-on exploratory data analysis, statistical modeling, feature engineering, model evaluation, and MLOps. Background spans engineering and physics simulation analytics, procurement data, and sensor/hyperspectral imaging, consistently turning model outputs into clear decisions for technical and non-technical teams.

PROFESSIONAL EXPERIENCE

University of Cincinnati

Research Assistant

Sep 2025 - Present

Cincinnati, OH

- Developed PyTorch and scikit-learn surrogate models for large-scale physics simulations, cutting runtime from 8 hours to sub-second predictions and screening 85% of candidate designs before full analysis.
- Built SQL and Python pipelines for feature engineering and model validation across 200+ source tables, applying anomaly checks, schema-drift detection, and data-quality rules that reduced out-of-specification records by 25%.
- Documented model performance, limitations, and findings in Power BI for technical and non-technical stakeholders.

Infosys

Data Scientist & Engineer, Analytics Platforms

May 2024 - Jul 2025

Remote

- Engineered ETL pipelines on AWS S3 and Parquet to ingest procurement, project, and vendor data from 30+ APIs, flat files, and web sources, standardizing schemas for downstream analytics and model development.
- Developed entity-resolution, deduplication, and data-profiling algorithms that reduced redundant records by 60% and improved data freshness by 40%, while surfacing trend, anomaly, and coverage insights for stakeholders.
- Developed GPT-4/LangChain extraction workflows for unstructured procurement documents, converting free-text records into structured features and reusable reporting outputs while maintaining validation rules and version-controlled documentation.

Hispec Lab, SRM University AP

Research Assistant

Jun 2022 - Jan 2024

Andhra Pradesh, India

- Delivered a production-grade HSI classification pipeline on EO-1 Hyperion satellite imagery - from raw sensor data through ENVI-based band removal, radiometric & atmospheric correction, to PPI/N-FINDR endmember detection producing a clean, analysis-ready dataset for the Krishna River Basin at scale.
- Eliminated a months-long manual labeling bottleneck by designing an unsupervised ground-truth generation system (LSTM + PCA), turning annotation from a 4+ month manual process into a fully automated pipeline.
- Improved accuracy and AUC-ROC by training and rigorously evaluating 25 deep learning and ensemble architectures; the final Stacking LSTM-CNN model decisively outperformed all standalone baselines.
- Accelerated model deployment decisions by benchmarking 8 feature engineering methods (CFS, RFE, LDA, MIFS vs. IPCA, PCA, QDA, NMF) across Fast-3D CNN and ResNet; identified IPCA as optimal on the Indian Pines and Salinas datasets.
- Solved a zero-labeled-data problem in medical AI by designing a synthetic data generation pipeline using Bilinear Mixing Models.

PROJECTS

NorthPeak Self-Serve Analytics Platform

- Built a dbt/Kimball star-schema warehouse over 3.3M+ e-commerce records (TheLook) with a governed single-source revenue ladder and a documented metric dictionary agreed with finance, replacing ad-hoc analyst requests with self-serve KPIs
- Enforced data quality with 73 dbt tests and 25 Great Expectations checks (caught 4/4 seeded anomalies), a daily Dagster refresh with fail-closed gating and alerts, and GitHub Actions CI on every PR.

AI-Powered Clinical Trial Match Engine

- Built an end-to-end RAG pipeline ingesting 50,000+ ClinicalTrials.gov records via semantic vector search and GPT-4 ranking, reducing manual trial-matching time from 4 hours to under 90 seconds per patient, achieving 81% top-3 accuracy validated against oncologist-reviewed ground truth on 500 cases.
- Deployed a production FastAPI microservice on AWS Lambda with async job-tracking, PostgreSQL result caching, and CloudWatch monitoring, handling 300+ requests/day at sub-3s response time, with domain-specific medical chunking improving retrieval precision by 34% over BM25 baseline.

TECHNICAL SKILLS

Python, SQL, R, ML & Statistics: scikit-learn, PyTorch, XGBoost, TensorFlow, statistical modeling, descriptive & inferential statistics, hypothesis testing, A/B testing, forecasting / time series, classification, clustering/segmentation, optimization, feature engineering, model evaluation, cross-validation, SHAP, MLflow, MLOps, model & data-drift monitoring, ETL/ELT pipelines, Databricks, Spark, Azure Data Lake, Snowflake, lakehouse architecture, Dagster, dbt, Great Expectations, GitHub Actions CI/CD, AWS (S3, Lambda), PostgreSQL, NLP & GenAI: RAG, LangChain, GPT-4, prompt engineering, Visualization & Collaboration: Power BI, Tableau, matplotlib, Seaborn, Agile/Scrum, Git, stakeholder communication.

PUBLICATIONS

- Efficient Hyperspectral Image Classification of the Krishna River Basin in Andhra Pradesh Using Hybrid Ensemble Learning Models in Optica Imaging Congress 2025, Optica Publishing Group..[Link](#)
- Hyperspectral Image Classification with Deep Learning: Unleashed by Feature Selection and Extraction in Innovations in Computer Science and Engineering, Springer Nature (2025).[Link](#)
- Non-Invasive Oral Cancer Detection Using Hyperspectral Imaging and Advanced Spectral Unmixing Models..[Link](#)

EDUCATION

University of Cincinnati

Master, Computer and Information Science

Aug 2025 - Apr 2027

Cincinnati, OH

SRM University AP

Bachelor of Technology, Computer Science(AI/ML)

Aug 2021 - May 2025

India

- GPA: 3.57/4.0

PATENTS & CERTIFICATIONS

- Patents (2): Automated Label Generation(IN202641066945 A1); Real-Time Food Contamination Detection (IN202441100125).
- OCI 2025 Certified Data Science Professional; OCI 2025 Certified Generative AI Professional